

Practical 1 Assignment

Print this document and bring it to the practical. You need to complete this assignment during the practical session.

After completing the form, log on to ClickUP and fill in your answers online (Practical 1).

Note: This form will not be collected, only the answers you submit to ClickUP will be evaluated.

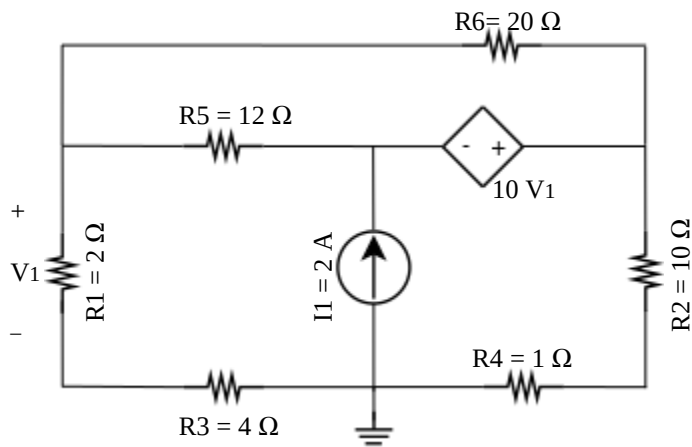
Important!

Use a fullstop to denote a decimal, do not use a comma, as follows:

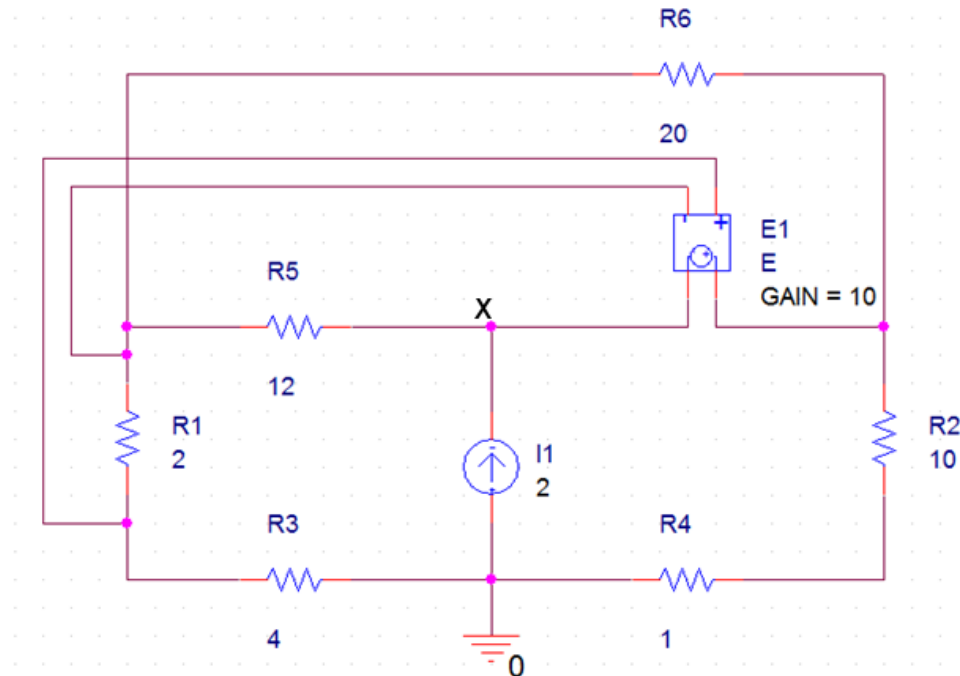
2.1 not 2,1

This is because clickUp cannot evaluate numbers using a comma. Also, do not include units (V, A, etc.) as clickUp cannot evaluate these either. Marks will be lost if these instructions are not followed.

Question 1



The Figure illustrates the common circuit diagram for Question 1.



This Figure shows the circuit diagram for capturing Question 1 in OrCAD.

Capture the above circuit schematic in OrCAD, and answer the questions that follow.

Hints:



If you struggle to find the DC current source, the part name is **I_{dc}**. Search for it under parts



If you struggle to find the voltage controlled voltage source, search for it by looking for the component named **E**. **Pay particular attention to the polarity of this element!**

Remember that the values underneath the resistors represent their resistances.

Questions:

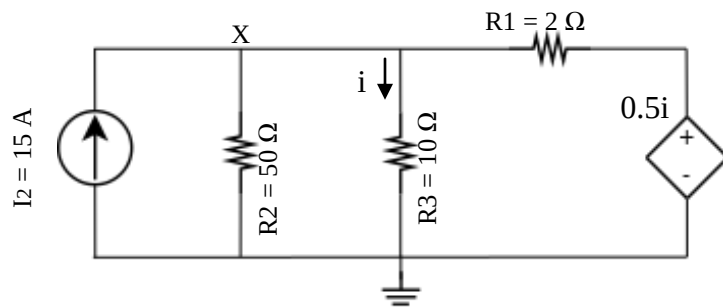
1.1) What is the voltage across R1?

1.2) What is the voltage across R6?

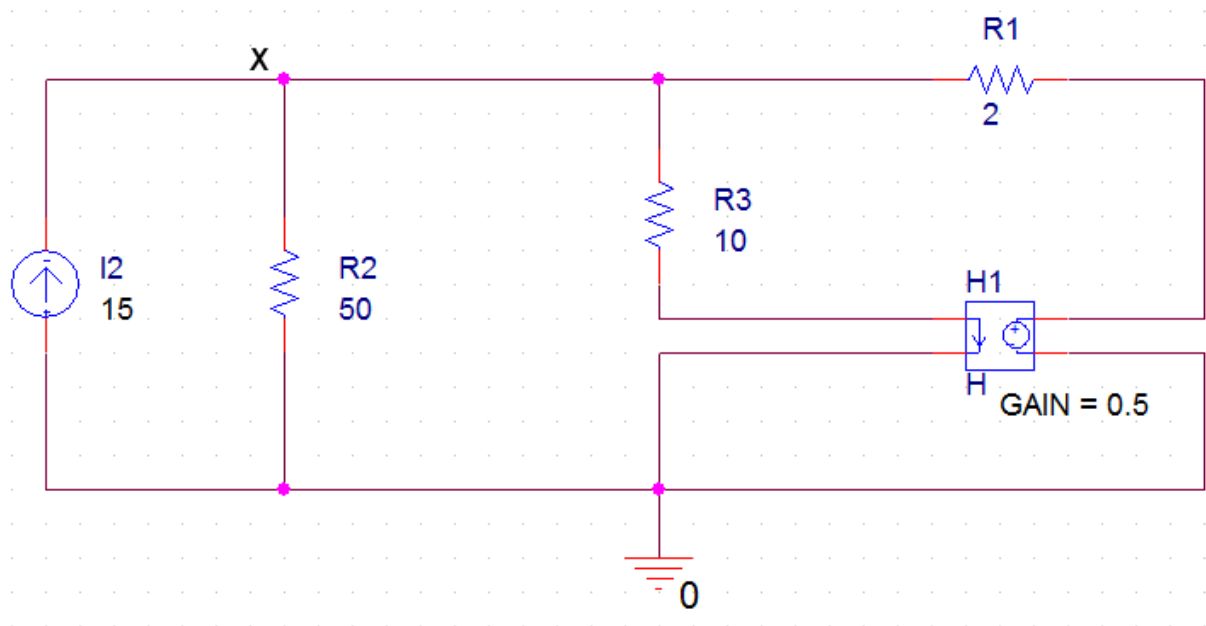
1.3) What is the current flowing through R2 and R4?

1.4) What is the node voltage at node X?

Question 2

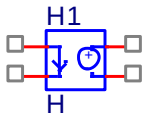


The Figure illustrates the common circuit diagram for Question 2.



This Figure shows the circuit diagram for capturing Question 2 in OrCAD.

Hints:



If you struggle to find the current controlled voltage source, search for it by looking for the component named **H**. **Pay particular attention to the polarity of this component!**

Questions:

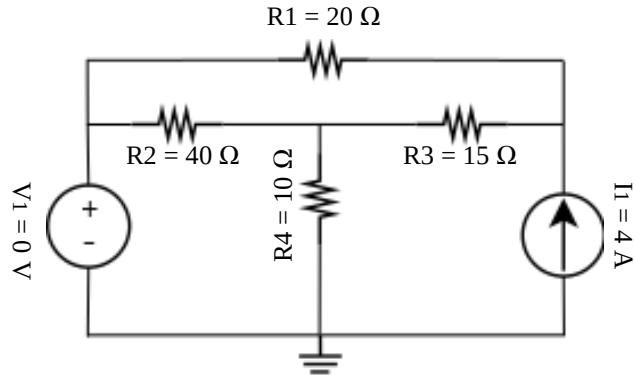
2.1) What is the node voltage at node X?

2.2) What is the current flowing through the $2\ \Omega$ resistor?

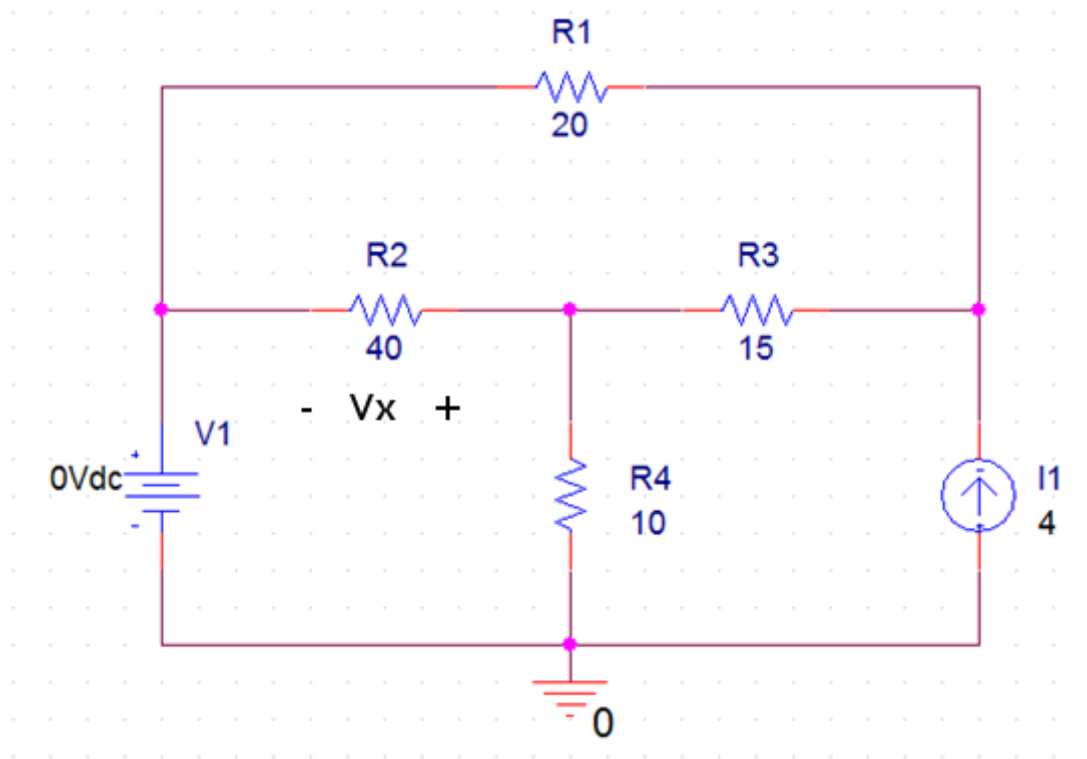
2.3) What is the total power dissipated?

2.4) What is the total power provided?

Question 3.1



The Figure illustrates the common circuit diagram for Question 3.



This Figure shows the circuit diagram for capturing Question 3 in OrCAD.

Create and simulate the circuit in figure 3 using the "DC Sweep" analysis type. Sweep $V1$ over the range 0 to 100 V in 20 V steps. Set $I1$ to 4A. Answer the questions that follow.

Hints:



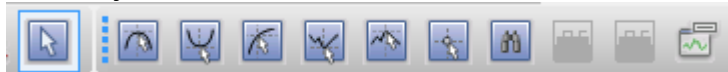
To display any signal after the DC sweep, select a probe and place it in the circuit where you would like the value to be read.

Questions:

3.1.1) What is the maximum current that flows through R1?

3.1.2) What is the minimum current that flows through R1?

Hints: To take accurate measurements from the graph, use the Cursor tool. In the PSpice window you should see these icons:

**Questions:**

3.1.3) What is the minimum value of the voltage V_{xx} ?

3.1.4) What is the maximum value of the voltage V_x ?

3.1.5) Complete the table below.

Voltage of V1 (Volt)	Voltage of V_x (Volt)
0	
20	
40	
60	
80	
100	

Hints:

Remember to use the correct voltage probes in order to measure voltage across a resistor. Also note the polarity of V_x (refer to figure 3).
Read the values of V_x off of the graph using the cursor tool.

Question 3.2

For the circuit created in question 3.1 (figure 3), set $V1 = 0$. Perform another DC sweep, but this time use the current source I1 as the sweep variable, and sweep it from 0 to 5 A in 1 A increments. Use a current probe to measure the current through R1.

3.2.1) Complete the table below. Provide answers correct to 2 decimal places.

Current I1 (Ampere)	I (R1) (Ampere)
0	
1	
2	
3	
4	
5	

Now log on to ClickUP and fill in your answers in:

PRACTICAL 1 ONLINE EVALUATION

Remember to use the decimal point and not the comma for number separation and do not supply units!